2026
Practical Parallel Computing
(実践的並列コンピューティング)

Part 1: OpenMP
No. 2: Sample Programs with Data Parallelism

Toshio Endo
endo@scrc.iir.isct.ac.jp

Overview of This Course

- Part 0: Introduction
  - 2 classes
- Part 1: Using OpenMP (OMP) ← We are here (2/4)
  - 4 classes
  - Report (required)
- Part 2: GPU programming with OpenACC (ACC)
  - 2 classes
  - Report (required)
- Part 3: GPU programming with CUDA
  - 3 classes
  - Report (elective)
- Part 4: Using MPI
  - 3 classes
  - Report (elective)

Summary of Previous Class

OpenMP is for shared-memory parallel programming

- #pragma omp parallel defines a parallel region, where multiple threads work simultaneously
- With #pragma omp for, loop-based programs can be parallelized easily
- Shared variables and private variables
- We have reviewed OpenMP version of mm sample

Today's Contents

Choose one of [O1]—[O4], and submit a report

Due date: May 7 (Thu)

[O1] Parallelize "diffusion" sample program by OpenMP.

[O2] Parallelize "bsort" sample program by OpenMP.

[O3] Parallelize "qsort" sample program by OpenMP.

[O4] (Freestyle) Parallelize *any* program by OpenMP.

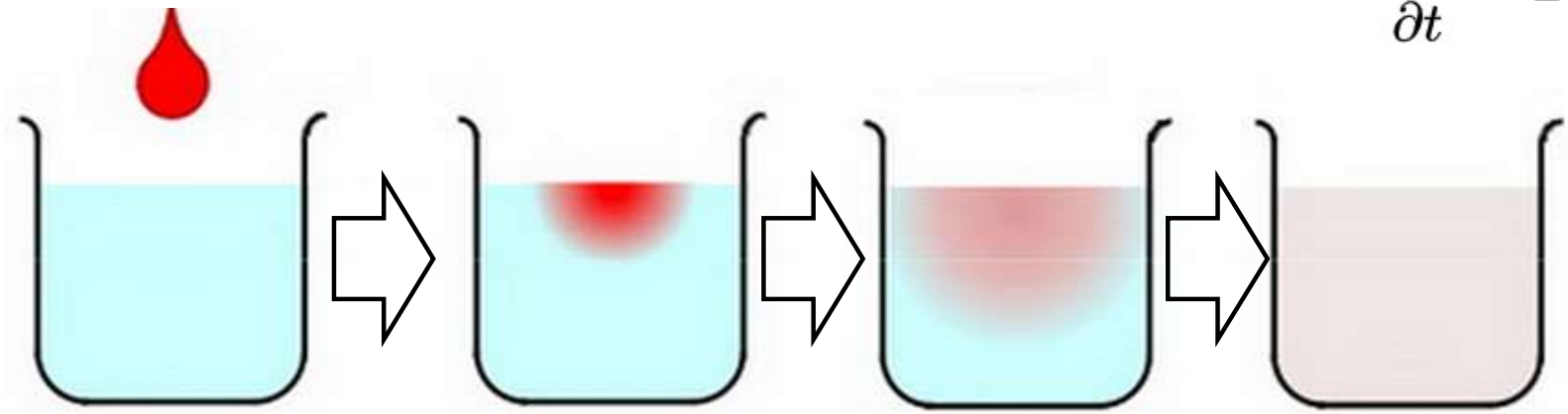
- Sample programs written with "for" loops are explained:
  - diffusion and bsort
- Sequential versions are at base/diffusion, base/bsort
  - There are already omp/diffusion, omp/bsort directories
  - They are only template and NOT parallelized
  - You can edit .c file there to solve [O1] or [O2]

“diffusion” Sample Program
（Target of [O1])

An example of diffusion phenomena:

- Pour a drop of ink into a water glass

$$\frac{\partial \phi}{\partial t} = D\nabla^2 \phi(\vec{r}, t)$$



The ink spreads gradually, and finally the density becomes uniform (Figure by Prof. T. Aoki, SCRC)

- Density of ink in each point vary according to time → Simulated by computers
  - cf) Weather forecast compute wind speed, temperature, air pressure…

“diffusion” Sample on TSUBAME

Available at ppcomp-ex/base/diffusion/
Go to the directory and do “make” command

- Execution : ./diffusion [nt]
- nt: Number of time steps
- nx, ny: Space grid size
  - nx=20000, ny=20000 (Fixed. See the code)
  - How can we make them variables? (mm sample will be useful as a reference)
- Compute Complexity : $O(nx \times ny \times nt)$

Notes about [O1]

You can use files in ppcomp-ex/omp/diffusion/

- diffusion.c is NOT parallelized; same as in base/diffusion/

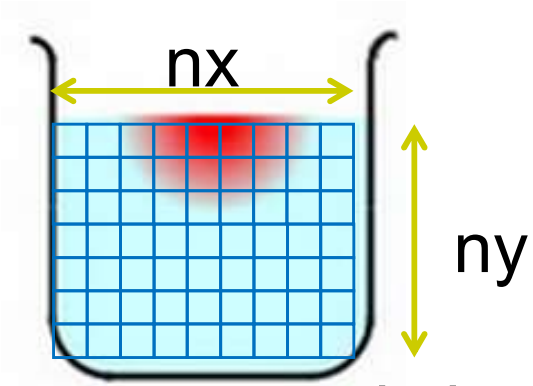
➔ Edit it for OpenMP parallelization

- Makefile uses “-fopenmp” option to support OpenMP

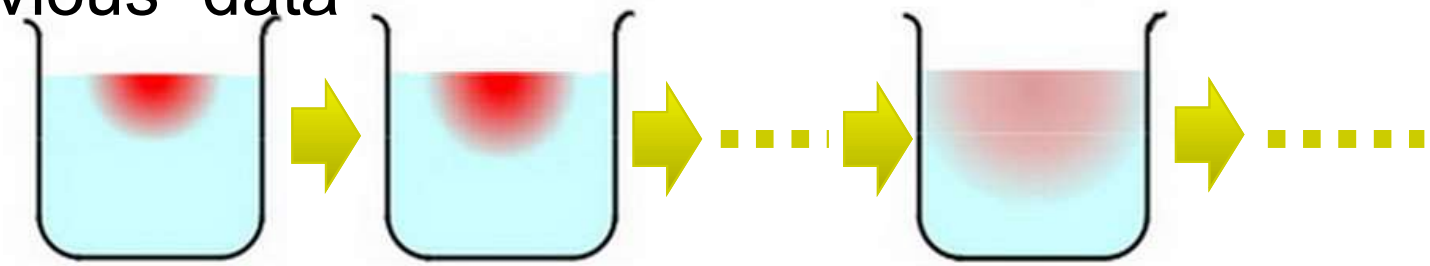
- omp/bsort, omp/qsort… have similar roles

Expression of Space to be Simulated

- Space to be simulated are divided into grids, and expressed by arrays (2D in this sample)



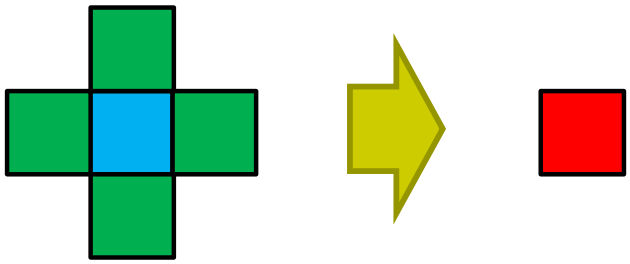
- Array elements are computed via timestep, by using "previous" data



Time step t=0 t=1 t=20

Stencil Computations

- In `diffusion', a data point *(x,y)* at time *t+1* is computed using following data
  - point *(x,y)* at time t
  - "Neighbor" points of (*x,y*) at time t



time t time t+1

  - In diffusion sample, the computation is simply "average of 5 points"
- Computations of similar type are called "stencil computations"
  - Frequently used in fluid simulations

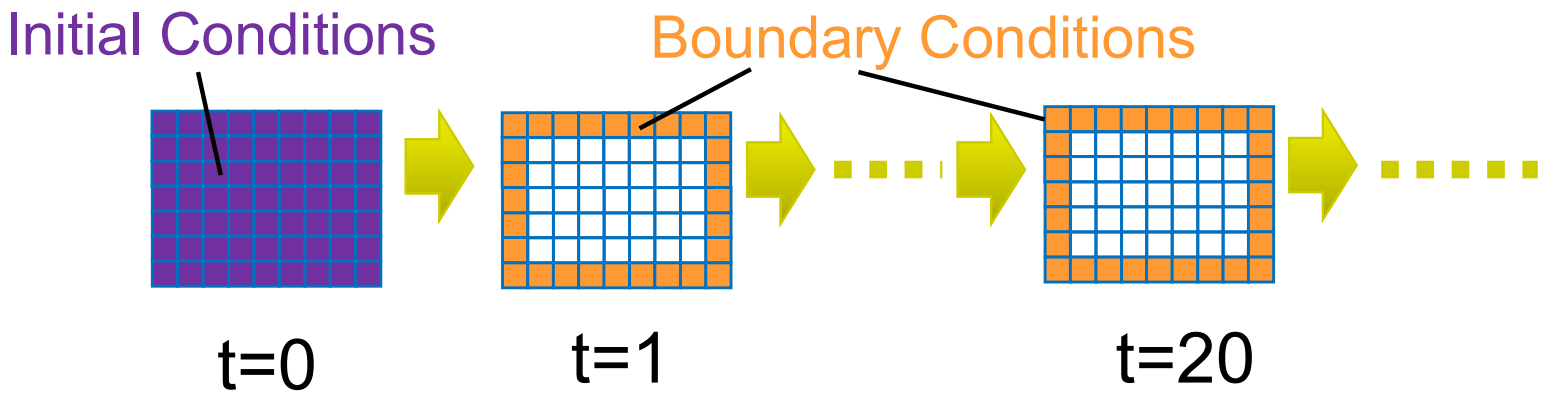
Original meanings of "stencil"

Initial Conditions & Boundary Conditions

In stencil computations, following data points cannot be computed

Instead, we have to give them (for example, as input data)

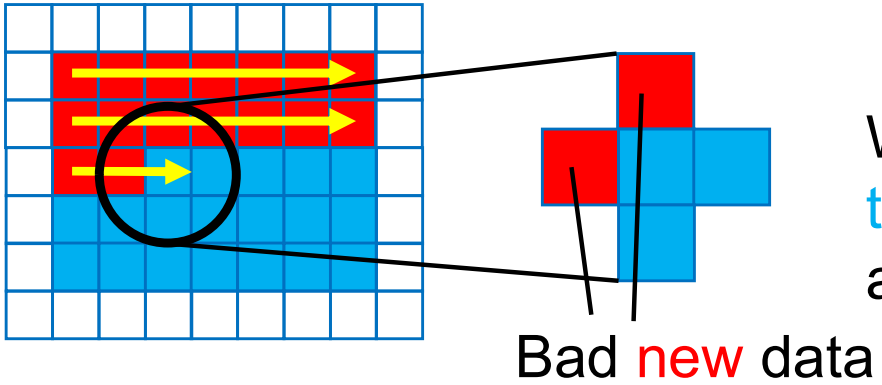
- All points at t=0 (Initial conditions)
  - In diffusion sample, given in init()
- "Boundary" points for all t (Boundary conditions)
  - In diffusion sample, they are constant during simulation
  - → See ranges of for-loops in calc(); boundaries are skipped
  - This is not good for simulation of a water glass ☹, but it's simple…



A Single Array Does not Work

Let us compute t ➔ t+1

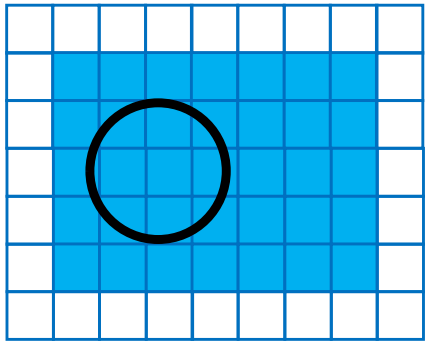
- With a single 2D array (Bug! ☹)



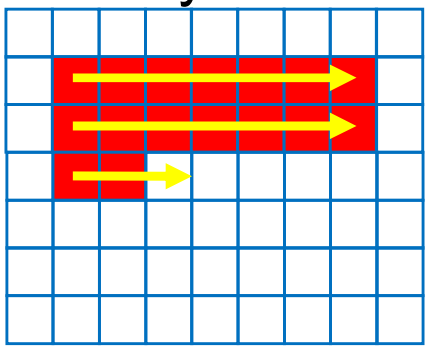
We need neighbor points at time t, but some have been already updated to t+1 ☹

- With separate 2D arrays (Good ☺)

An array for t



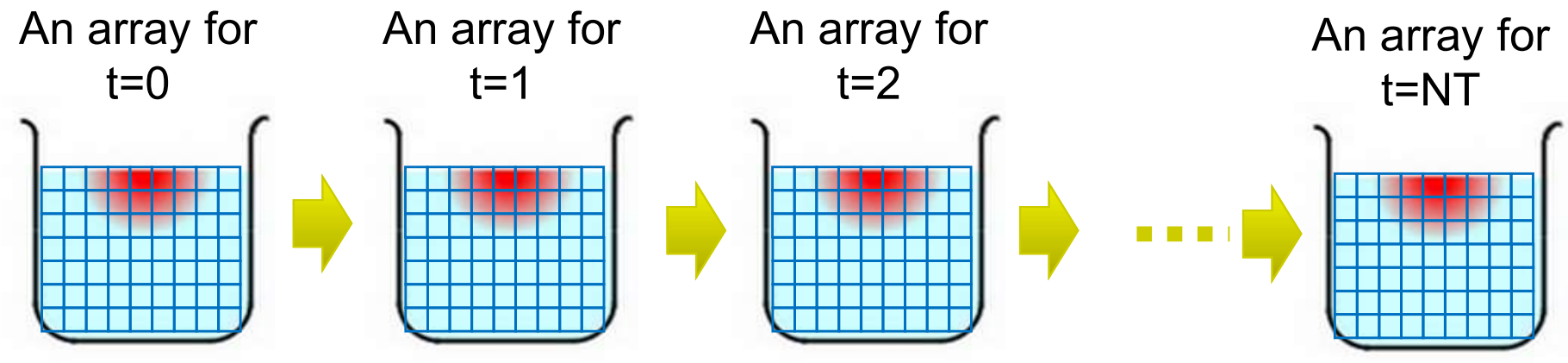
An array for t+1



We can access "old" neighbor points correctly ☺

Multiple Arrays?

- We repeat update of the array for NT times



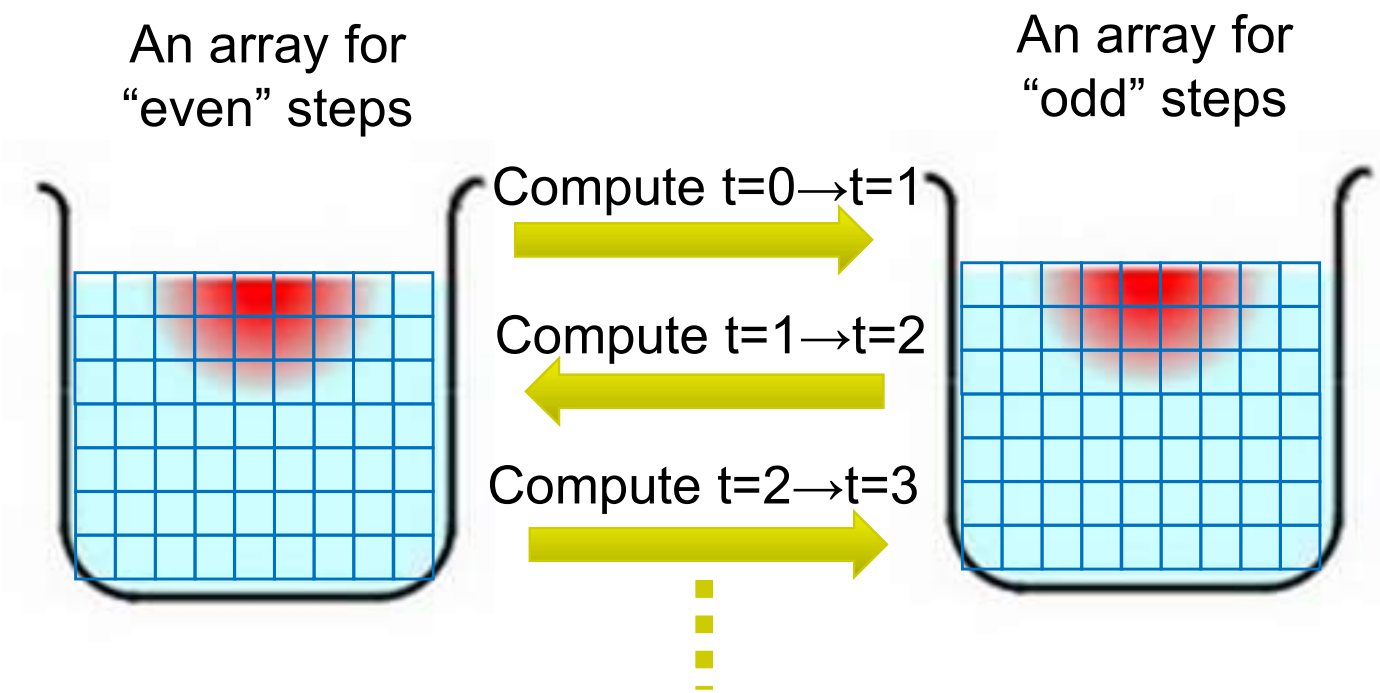
A simple way is to make arrays for all time steps

```
float dens[NT+1][NY][NX]
```

- This uses too much memory
- Do we need all of (NT+1) arrays?

Double Buffering Technique

- It is sufficient to have “current” array and “next” array.
- It is better to use only “Double buffers”



The diffusion sample program uses

```
float dens[2][NY][NX];
```

Notes about 3D Arrays

```
#define NX 15000
#define NY 15000

float dens[2][NY][NX];
```

Is this a variable length array? ➔ No.

With #define syntax, the above is compiled as below

```
float dens[2][15000][15000];
```

This is a fixed size variable, so NX/NY cannot be changed after compilation. (On the other hand, `mm' can do that)

※ This is related to an optional topic in [O1]

How Do We Parallelize "diffusion"?

calc() takes long time, complexity is O(nx ny nt)

It mainly uses "for" loops

➔ #pragma omp parallel for is useful! But…

There are 3 (t, x, y) loops. Which should be parallelized?

[Hint1] Parallelizing either of spatial loop (x, y) would be good. Then spaces are divided into multiple threads

→ [Q] Parallelizing t loop is a not good idea. Why?

[Hint2] Take care of "pitfall in nested loops" (see slides in previous class)

Towards "Correct" Parallel Programming

There are several types of bugs in parallel programming

- Bugs in compile time

- Bugs in run time
  - Bugs that abort execution (cf. segmentation fault)
  - Silent bugs → Hardest to find!

All bugs should be avoided!

When Can We Use "omp for"?

- Loops with some (complex) forms cannot be supported, unfortunately ☹
- The target loop must be in the following form

```
#pragma omp for
  for (i = value; i op value; incr-part)
    body
```

"*op*" : <, >, <=, >=, etc.
"*incr-part*" : i++, i--, i+=c, i-=c, etc.

OK ☺: `for (x = n; x >= 0; x-=4) …`

ERROR ☹: `for (i = 0; test(i); i++) …`

ERROR ☹: `for (p = head; p != NULL; p = p->next)`

Errors in compile time

What are Differences between These Codes?

```
double D[100];
        :
```

Code A

```
#pragma omp parallel for
  for (i = 0; i < 100; i++) {
     D[i] = D[i]+1.0;
  }
```

Code B

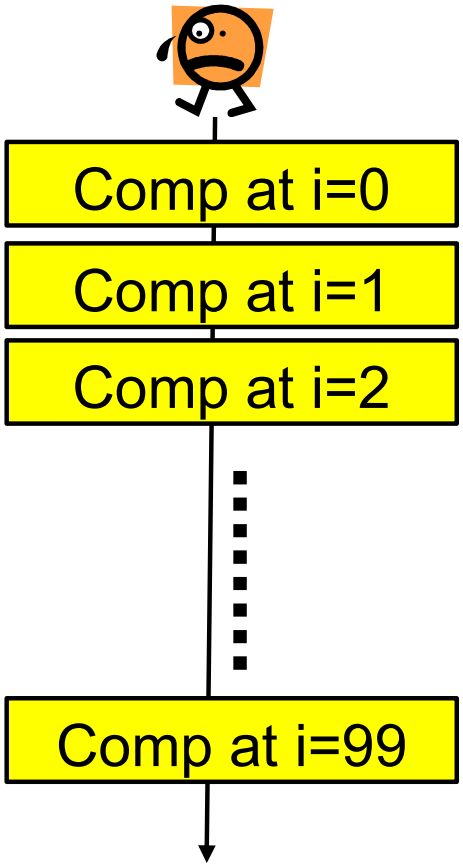
```
#pragma omp parallel for
  for (i = 0; i < 99; i++) {
     D[i+1] = D[i]+1.0;
  }
```

- Both codes can be compiled and executed…
- But only code A is correct ☺ , code B has a bug ☹
  - Code B's results may be wrong

Sequential Execution and Parallel Execution of Loop

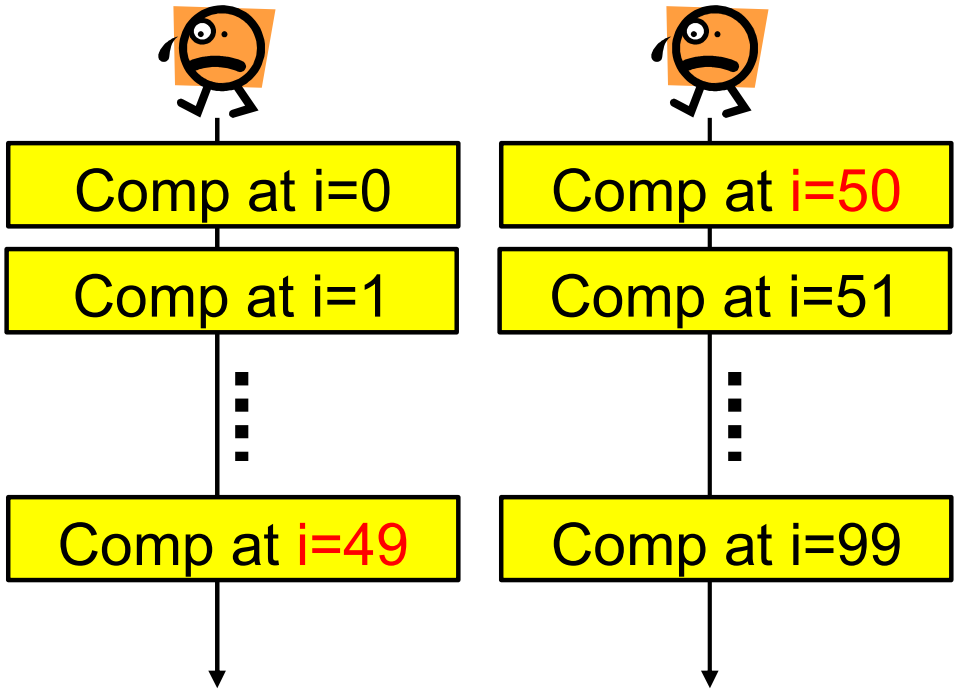
[Sequential]

```
for (i = 0; i < 100; i++) …
```



[Parallel]

```
#pragma omp parallel for
  for (i = 0; i < 100; i++) …
```



in case of 2 threads,
i=50 is computed before i=49

Difference between Two Codes

Code A

```
#pragma omp parallel for
  for (i = 0; i < 100; i++) {
     D[i] = D[i]+1.0;
  }
```

OK

It is ok to reorder 100 computations

Code B

```
#pragma omp parallel for
  for (i = 0; i < 99; i++) {
     D[i+1] = D[i]+1.0;
  }
```

NG

Computations must be done in an order (i=0,1,2…)

➔ Parallelization breaks the order

Dependency between Computations

We define following sets for computation C

- Read set R(C): the set of variables read by C
- Write set W(C): the set of variables written by C
  - Ex) C: x = y+z ➔ R(C) = {y, z}, W(C) = {x}

We define dependency between C1 and C2

- If (W(C1) ∩ R(C2) ≠ ∅), C1 and C2 are dependent (write vs read)
- If (R(C1) ∩ W(C2) ≠ ∅), C1 and C2 are dependent (read vs write)
- If (W(C1) ∩ W(C2) ≠ ∅), C1 and C2 are dependent (write vs write)
- Otherwise, C1 and C2 are independent
  - ※ read vs read cases are independent

<u>If C1 and C2 are independent, parallelization of C1 and C2 is safe ☺</u>

Example of Dependency

Code A

```
#pragma omp parallel for
  for (i = 0; i < 100; i++) {
     D[i] = D[i]+1.0;    ← A_i
  }
```

$R(A_i) = \{D[i]\}$, $W(A_i) = \{D[i]\}$

All 100 computations are independent

Code B

```
#pragma omp parallel for
  for (i = 0; i < 99; i++) {
     D[i+1] = D[i]+1.0;  ← B_i
  }
```

$R(B_i) = \{D[i]\}$, $W(B_i) = \{D[i+1]\}$

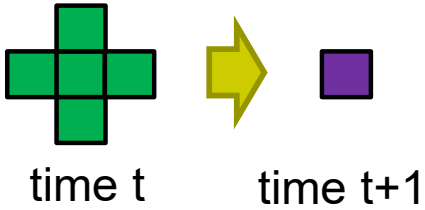
$R(B_{i+1}) \cap W(B_i) = \{D[i+1]\} \neq \emptyset$ → Dependent!

Dependency and Parallelism in Stencil Computations

Consider 1D stencil computation:

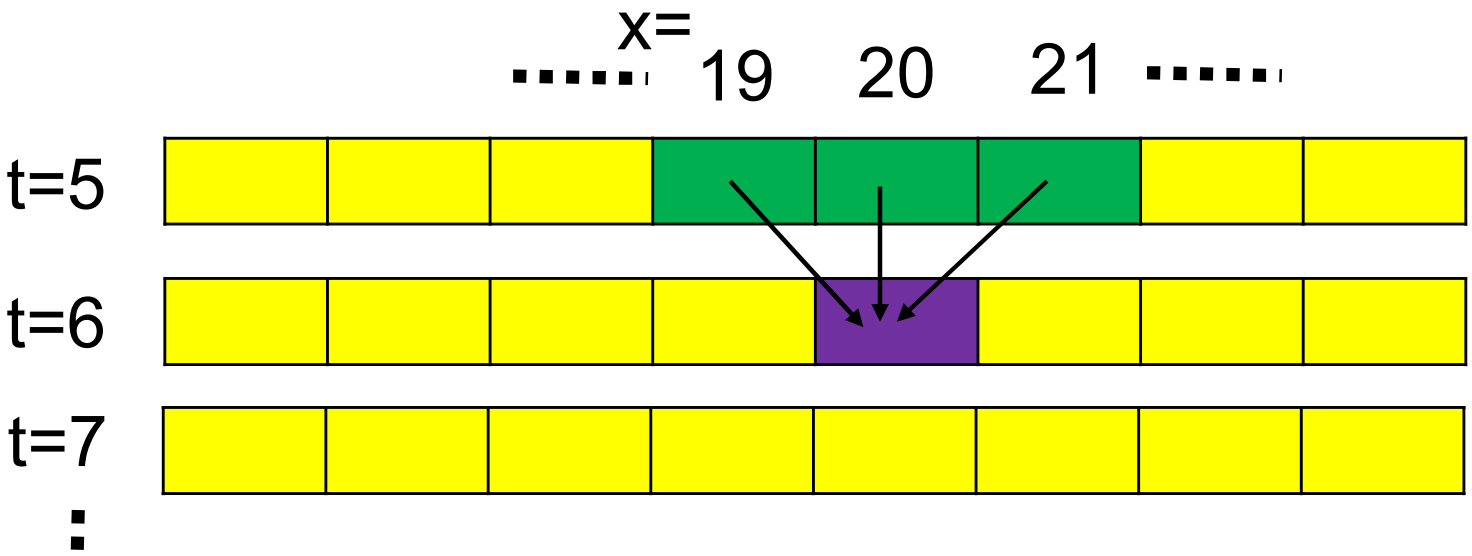
```
for (t = 0; t < NT; t++)
    for (x = 1; x < NX-1; x++)
        f_{t+1,x} = (f_{t,x-1} + f_{t,x} + f_{t,x+1}) / 3.0   /*  c_{t,x} */
```

※ This is simpler than "diffusion" (2D) sample



We let $c_{t,x}$ be computation of a single point $f_{t+1,x}$

$R(c_{t,x}) = \{f_{t,x-1}, f_{t,x}, f_{t,x+1}\}$, $W(c_{t,x}) = \{f_{t+1,x}\}$



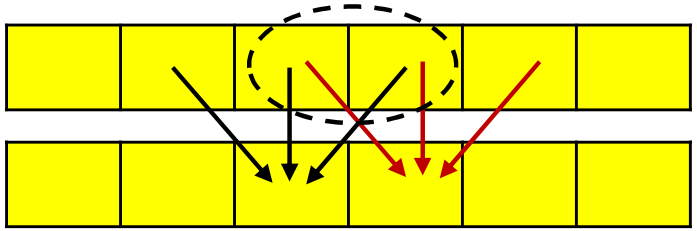
※ This figure omits double buffering technique

Discussion on Stencil: Case of Spatial Loop

```
for (t = 0; t < NT; t++)
    for (x = 1; x < NX-1; x++)        ← Is this loop parallelizable?
        f_{t+1,x} = (f_{t,x-1} + f_{t,x} + f_{t,x+1}) / 3.0   /*  c_{t,x} */
```

- Can we compute $c_{5,20}$ and $c_{5,21}$ in parallel? *(t is same, x is different)*
  - $R(C_{5,20})=\{f_{5,19}, f_{5,20}, f_{5,21}\}$, $W(C_{5,20})=\{f_{6,20}\}$
  - $R(C_{5,21})=\{f_{5,20}, f_{5,21}, f_{5,22}\}$, $W(C_{5,21})=\{f_{6,21}\}$
  - → They are independent ☺ ,for all pairs of x
  - → x loop can be parallelized

Read vs. Read is Ok



Discussion on Stencil: Case of Temporal Loop

for (t = 0; t < NT; t++) ← Is this loop parallelizable?

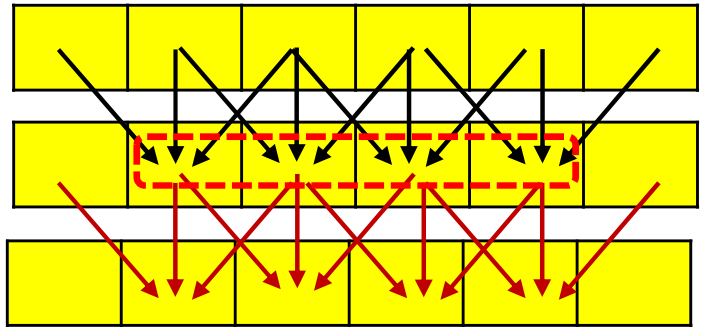
```
for (x = 1; x < NX-1; x++)
    f_{t+1,x} = (f_{t,x-1} + f_{t,x} + f_{t,x+1}) / 3.0
```

C_t

- Can we compute C_5 and C_6 in parallel? *(t is different)*
  - $R(C_5)=\{f_{5,0}, \ldots, f_{5,NX-1}\}$, $W(C_5)=\{f_{6,1}, \ldots, f_{6,NX-2}\}$
  - $R(C_6)=\{f_{6,0}, \ldots, f_{6,NX-1}\}$, $W(C_6)=\{f_{7,1}, \ldots, f_{7,NX-2}\}$

→ $R(C_6) \cap W(C_5) = \{f_{6,1}, \ldots, f_{6,NX-2}\} \neq \emptyset$

→ They are dependent ☹



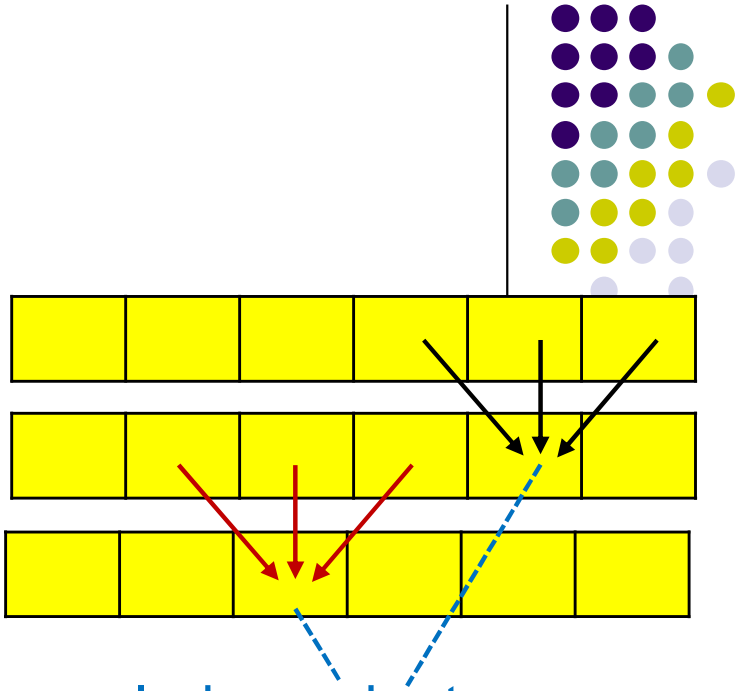
dependent!!

In Assignment [O1]
- it is OK to parallelize x-loop or y-loop
- it is NG to parallelize t-loop

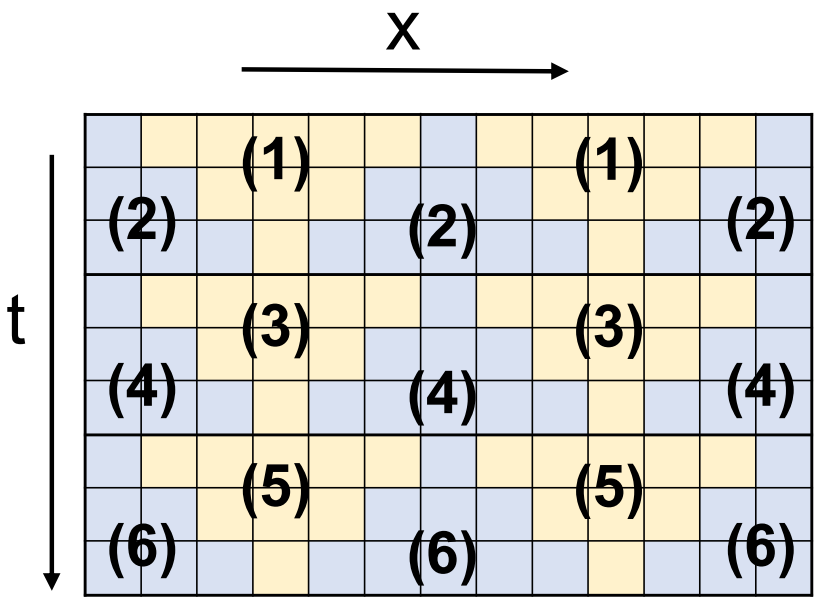
Advanced Topic: More Speed in Stencil

We see dependency more in detail:

- $c_{6,20}$ depends on $c_{5,19}$, $c_{5,20}$, $c_{5,21}$
  - The same point or its direct neighbor
- But not on $c_{5,22}$



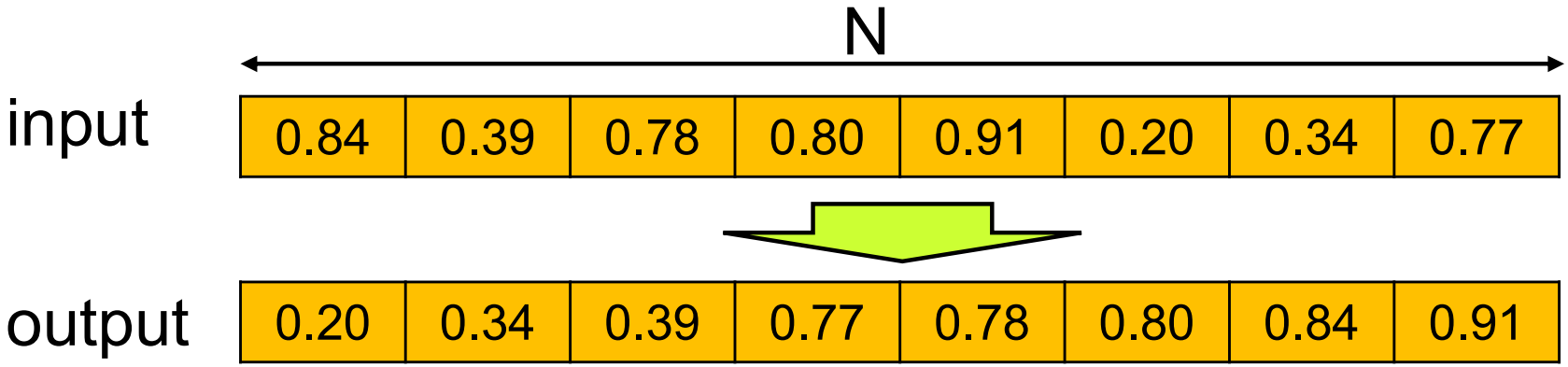
Independent
➔ Can be parallelized
➔ Can be reordered



Temporal blocking technique:
After computations in (1) finish, we can start (2)
"Trapezoids" in the same stage can be parallelized
➔ Speed is improved for better access locality

“bsort” Sample Program （Target of [O2]）

- Sort an array with bitonic sort algorithm



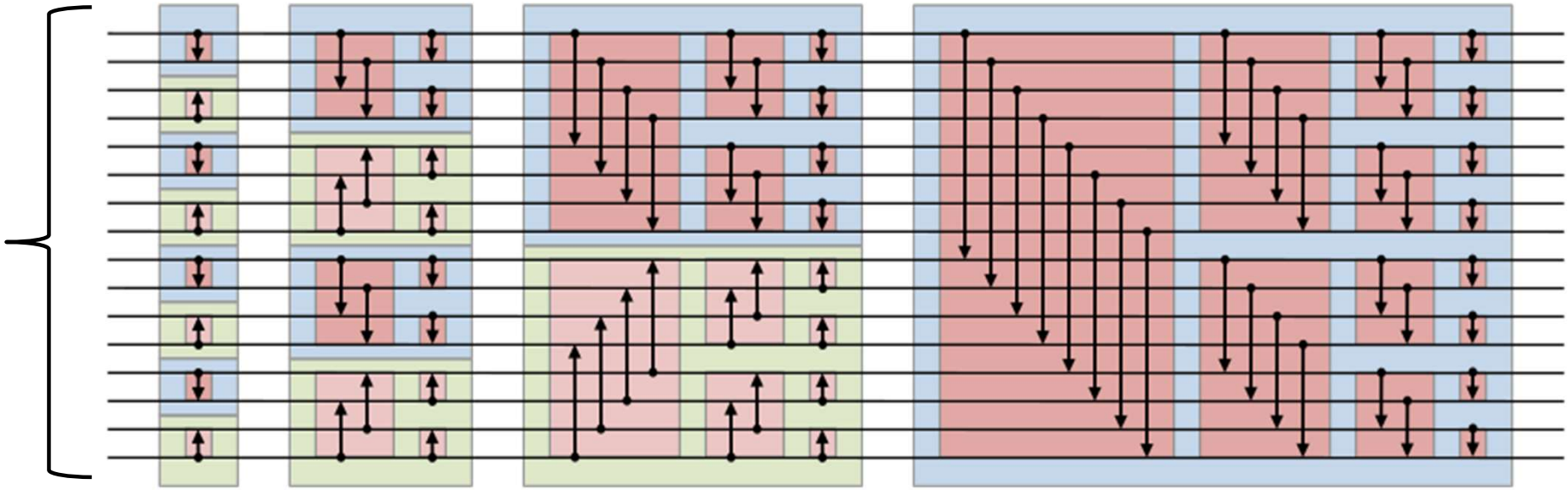
In output, for all i, data[i] ≦ data[i+1]

- There are many famous sorting algorithm with different compute complexity:
  - bubble sort and insertion sort, $O(N^2)$
  - bitonic sort [O2], $O(N \log^2 N) = O(N (\log N)^2)$
  - quick sort [O3], $O(N \log N)$
  - merge sort, $O(N \log N)$
  - radix sort

Overview of bitonic sort

Case of N=16 (2^4)

time →

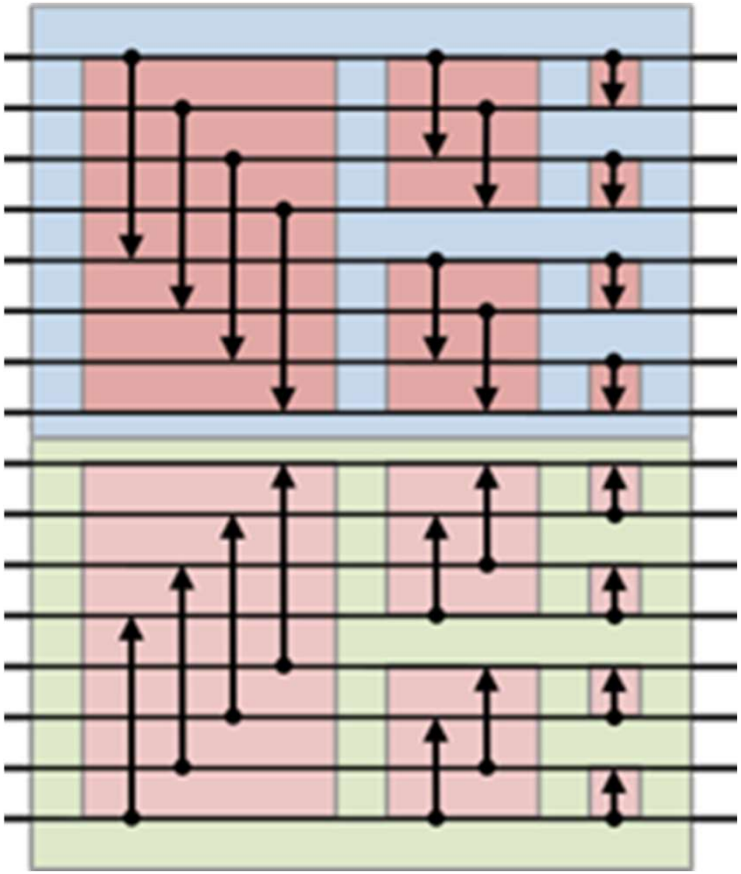


log N steps

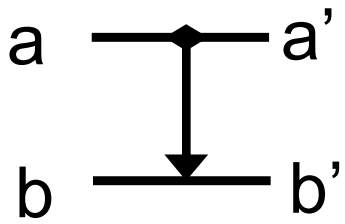
the figure is from Wikipedia

Basic Operations in bsort

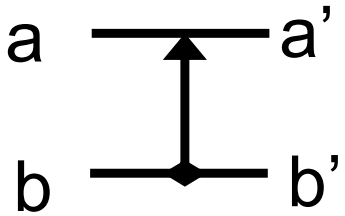
Step *k*



sub-step sub-step sub-step



if a≦b, a'=a, b'=b
if a>b, a'=b, b'=a



if a≦b, a'=b, b'=a
if a>b, a'=a, b'=b

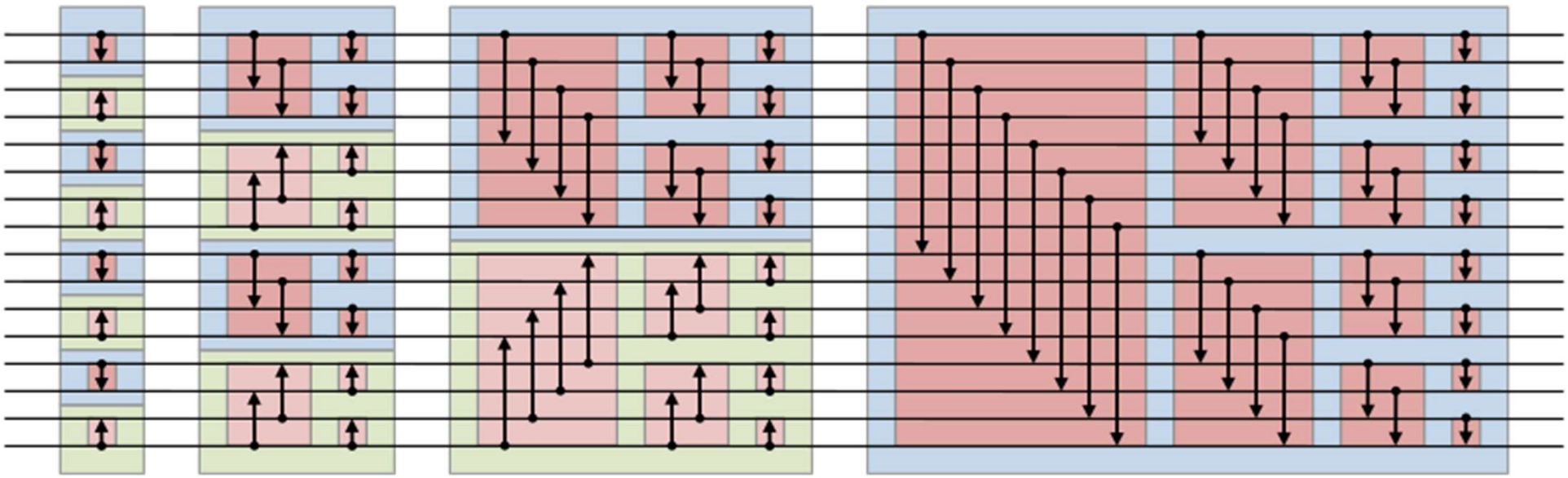
After blue box, data are sorted in ascending order

After green box, data are sorted in descending order

if interested in its proof, please try Google/ChatGPT

Computation Complexity

Sub-step: O(N) ➔ Step k: O(kN)



step 1 step 2 step 3 step 4 = S = log N

Total step number S = log N

Total complexity: $O(N + 2N + 3N + ... + S\,N)$

$= O(S^2\,N) = O(N \log^2 N)$

Loops in bsort

- Bitonic sort assumes number of data is power of 2
- We compute N2 ≧ N, N2 is power of 2, and copy data

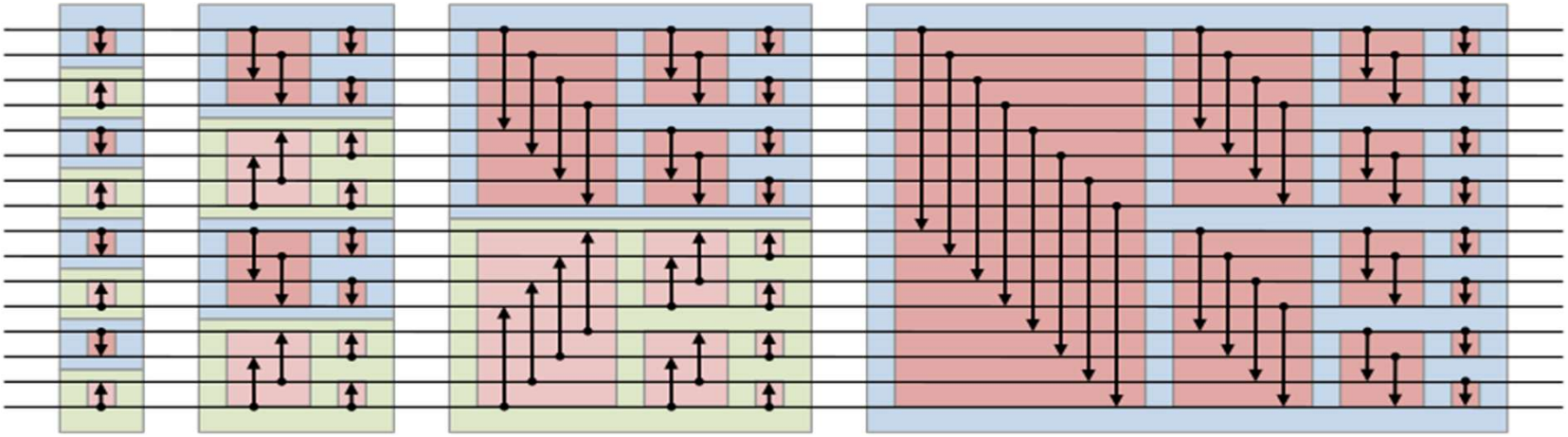
loop structure of sort()

1<<i = 2^i. (<< is bit-shift)

```
for (i = 1; (1<<i) <= N2; i++) {  // step loop
  for (j = i-1; j >= 0; j--) {    // sub-step loop
      :
      for (k = 0; k < N2; k++) {
          // compare 2 data and swap if appropriate
      }
  }
}
```

Program is a bit complicated for bit operations,
but you can use as it is ☺

How Do We Parallelize "bsort"?



step 1 step 2 step 3 step 4 = (log N)

- Since steps/sub-steps cannot be reordered, we should NOT parallelize *i, j* loops
- Please confirm that parallelizing *k* loops is safe

Why Do We Use both bsort and qsort?

- Quick sort with O(N log N) complexity is usually faster than bitonic sort with O(N $\log^2$ N)
- Bitonic sort is easier for parallelization
  - In OpenMP, we can use #pragma omp for
    - OpenMP version of qsort will be explained in next class
  - Also parallelizing bsort with OpenACC/CUDA/MPI is easier
    - qsort is harder, and out of scope in this class ☹
    - cf) Cederman et al. “GPU-Quicksort: A practical Quicksort algorithm for graphics processors”, JEA vol14, 2010

Assignments in OpenMP Part (Abstract)

Choose one of [O1]—[O4], and submit a report

Due date: May 7 (Thu)

[O1] Parallelize “diffusion” sample program by OpenMP.

[O2] Parallelize “bsort” sample program by OpenMP.

[O3] Parallelize “qsort” sample program by OpenMP.

[O4] (Freestyle) Parallelize *any* program by OpenMP.

For more detail, please see ppcomp26-1-1 slides

Plan of Part 1 (OMP)

- No. 1
  - Introduction to OpenMP
- No. 2 (Today)
  - Data parallelism with for loops
  - diffusion sample [O1], bsort sample [O2]
- No. 3
  - Task parallelism
  - qsort sample [O3]
- No. 4
  - What are bottlenecks, race conditions?